

Mathematics for Mechatronics Engineering

Laplace Transforms and Engineering Applications

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- **Session 1:** Foundations of Laplace Transforms
- **Session 2:** Inverse Laplace and Partial Fractions
- **Session 3:** Transfer Functions and Dynamic Systems

Goal: Use Laplace transforms to analyze dynamic engineering systems.

Why Engineers Use Laplace Transforms

Differential equations are difficult to solve directly.

Example:

$$m\ddot{x} + b\dot{x} + kx = f(t)$$

Key Idea

Laplace transforms convert differential equations into algebra equations.

This simplifies analysis of:

- Motors
- Circuits
- Vibrations
- Control systems

Definition of Laplace Transform

The Laplace transform of $f(t)$ is:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

Interpretation

The Laplace transform converts a time-domain signal into the s -domain.

Common Laplace Transforms

$$\mathcal{L}\{1\} = \frac{1}{s}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\mathcal{L}\{\sin \omega t\} = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$$

Transforming Derivatives

This is the most important property.

$$\mathcal{L} \left\{ \frac{df}{dt} \right\} = sF(s) - f(0)$$

Second derivative:

$$\mathcal{L} \left\{ \frac{d^2f}{dt^2} \right\} = s^2F(s) - sf(0) - f'(0)$$

Important

Initial conditions appear automatically.

Solving a Differential Equation

Example:

$$\frac{dx}{dt} + 2x = 0$$

with:

$$x(0) = 3$$

Take Laplace transform:

$$sX(s) - 3 + 2X(s) = 0$$

$$(s + 2)X(s) = 3$$

$$X(s) = \frac{3}{s + 2}$$

Inverse Transform

Using transform tables:

$$X(s) = \frac{3}{s+2}$$

Inverse Laplace transform:

$$x(t) = 3e^{-2t}$$

Physical Meaning

The response decays exponentially over time.

RC circuit equation:

$$RC \frac{dv}{dt} + v = V_s$$

Laplace transforms simplify transient analysis.

Applications:

- Charging capacitors
- Motor response
- Sensor dynamics

Laplace transform in MATLAB:

```
syms t s  
laplace(exp(-2*t))
```

Output:

$$\frac{1}{s + 2}$$

Practice Exercise

Find the Laplace transform of:

$$f(t) = 5e^{-3t}$$

Then solve:

$$\frac{dx}{dt} + x = 0$$

with:

$$x(0) = 2$$

Why Laplace Transform is Powerful

Time-domain differential equation:

$$\frac{dx}{dt} + 2x = 5$$

requires calculus techniques.

Laplace transform converts it into:

$$sX(s) - x(0) + 2X(s) = \frac{5}{s}$$

which is algebra.

Key Insight

Laplace transform converts differentiation into multiplication by s .

Important Transform Table

$$\mathcal{L}\{1\} = \frac{1}{s}$$

$$\mathcal{L}\{e^{-at}\} = \frac{1}{s+a}$$

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$$

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$

MATLAB Example: Exponential Decay

```
t = 0:0.01:10;  
x = 3*exp(-2*t);  
plot(t,x)  
xlabel('Time')  
ylabel('x(t)')
```

Observation

The response decays exponentially toward zero.

Modify the decay rate:

$$x(t) = 3e^{-at}$$

Try:

$$a = 0.5, \quad 2, \quad 5$$

Observe:

- Faster decay
- Slower decay
- System response speed

Inverse Laplace Transform

Inverse Laplace transform converts from s -domain back to time domain.

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

Example:

$$\mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\} = e^{-3t}$$

Why Partial Fractions Matter

Many Laplace expressions are complicated.

Example:

$$\frac{5}{(s+1)(s+2)}$$

Goal

Rewrite into simpler fractions that match transform tables.

Partial Fraction Decomposition

Assume:

$$\frac{5}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

Multiply both sides:

$$5 = A(s+2) + B(s+1)$$

Solve for A, B .

Solving for Coefficients

Expand:

$$5 = As + 2A + Bs + B$$

Group terms:

$$5 = (A + B)s + (2A + B)$$

Compare coefficients:

$$A + B = 0$$

$$2A + B = 5$$

Solution:

$$A = 5, \quad B = -5$$

Therefore:

$$\frac{5}{(s+1)(s+2)} = \frac{5}{s+1} - \frac{5}{s+2}$$

Inverse transform:

$$f(t) = 5e^{-t} - 5e^{-2t}$$

Unit Step Function

The unit step function models switching behavior.

$$u(t - a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

Transform:

$$\mathcal{L}\{u(t - a)\} = \frac{e^{-as}}{s}$$

Engineering Application: Switching Circuit

Example:

- Voltage source turns on at $t = 2$
- Represented using unit step functions

$$v(t) = 5u(t - 2)$$

Applications

- Digital control
- Switching electronics
- Delayed inputs

RC Circuit Example

Circuit equation:

$$RC \frac{dv}{dt} + v = V_s$$

Laplace transform:

$$RC(sV(s) - v(0)) + V(s) = \frac{V_s}{s}$$

Solve algebraically for $V(s)$.

Inverse Laplace transform in MATLAB:

```
syms s t  
ilaplace(1/(s+2))
```

Output:

$$e^{-2t}$$

Practice Exercise

Find the inverse Laplace transform:

$$F(s) = \frac{4}{(s+1)(s+3)}$$

Use:

- 1 Partial fractions
- 2 Transform table

Worked Example

Find:

$$\mathcal{L}^{-1} \left\{ \frac{2s + 5}{(s + 1)(s + 2)} \right\}$$

Assume:

$$\frac{2s + 5}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2}$$

Solve for A and B .

Repeated Poles

Example:

$$\frac{1}{(s+1)^2}$$

Inverse transform:

$$te^{-t}$$

Important

Repeated poles create slower responses.

MATLAB Example: RC Circuit

```
t = 0:0.01:5;  
v = 1-exp(-t);  
plot(t,v)
```

Observation

Capacitor voltage gradually approaches steady state.

Plot:

$$v(t) = 1 - e^{-t/\tau}$$

for:

$$\tau = 0.5, \quad 1, \quad 3$$

Observe:

- Fast charging
- Slow charging
- Time constant effect

Second-Order Systems

Many mechatronic systems are second-order systems.

Example:

$$m\ddot{x} + b\dot{x} + kx = f(t)$$

- m : mass
- b : damping
- k : stiffness

Laplace Transform of Second-Order Systems

Taking Laplace transform:

$$m(s^2X(s) - sx(0) - x'(0)) + b(sX(s) - x(0)) + kX(s) = F(s)$$

Key Advantage

Differential equations become algebra equations.

Transfer Functions

Transfer function:

$$G(s) = \frac{\text{Output}}{\text{Input}}$$

Example:

$$G(s) = \frac{1}{ms^2 + bs + k}$$

Meaning

Transfer functions describe system behavior in the s -domain.

Poles and Stability

Poles are roots of the denominator.

Example:

$$G(s) = \frac{1}{s^2 + 2s + 5}$$

Poles:

$$-1 \pm 2j$$

Interpretation

Poles determine:

- Stability
- Oscillation
- Damping

Types of Responses

Overdamped

Two real negative poles.

Critically Damped

Fastest response without oscillation.

Underdamped

Complex poles create oscillation.

Block Diagrams

Control systems are often represented using block diagrams.

Typical blocks:

- Controller
- Plant
- Sensor
- Feedback

Why Important

Block diagrams simplify analysis of complex systems.

Electrical Engineering Example

RLC circuit:

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = v(t)$$

Laplace transform:

$$(Ls^2 + Rs + \frac{1}{C})Q(s) = V(s)$$

Applications

- Filters
- Resonance
- Power electronics

Step response in MATLAB:

```
num = [1];  
den = [1 2 5];  
step(tf(num,den))
```

Practice Exercise

Given:

$$G(s) = \frac{1}{s^2 + 4s + 8}$$

Find:

- ① Poles
- ② Stability
- ③ Type of response

Mass-Spring-Damper System

$$m\ddot{x} + b\dot{x} + kx = f(t)$$

- Mass resists acceleration
- Damper dissipates energy
- Spring restores position

Applications

- Robot joints
- Vehicle suspension
- Motor systems

MATLAB Example: Underdamped System

```
t = 0:0.01:10;  
x = exp(-0.2*t).*cos(3*t);  
plot(t,x)
```

Observation

Oscillation gradually decreases because of damping.

MATLAB Exercise: Damping Comparison

Compare:

$$x(t) = e^{-at} \cos(3t)$$

for:

$$a = 0, \quad 0.2, \quad 1$$

Observe:

- No damping
- Light damping
- Heavy damping

Pole Locations and Stability

Pole locations determine system behavior.

- Left-half plane: stable
- Right-half plane: unstable
- Imaginary axis: sustained oscillation

Key Insight

Poles are closely related to eigenvalues.

```
num = [1];  
den = [1 2 5];  
G = tf(num,den);  
step(G)
```

Result

Visualize the system step response directly.

Summary

- Laplace transforms simplify differential equations
- Partial fractions simplify inverse transforms
- Transfer functions describe system dynamics
- Poles determine stability and oscillation